HÓA HỮU CƠ A

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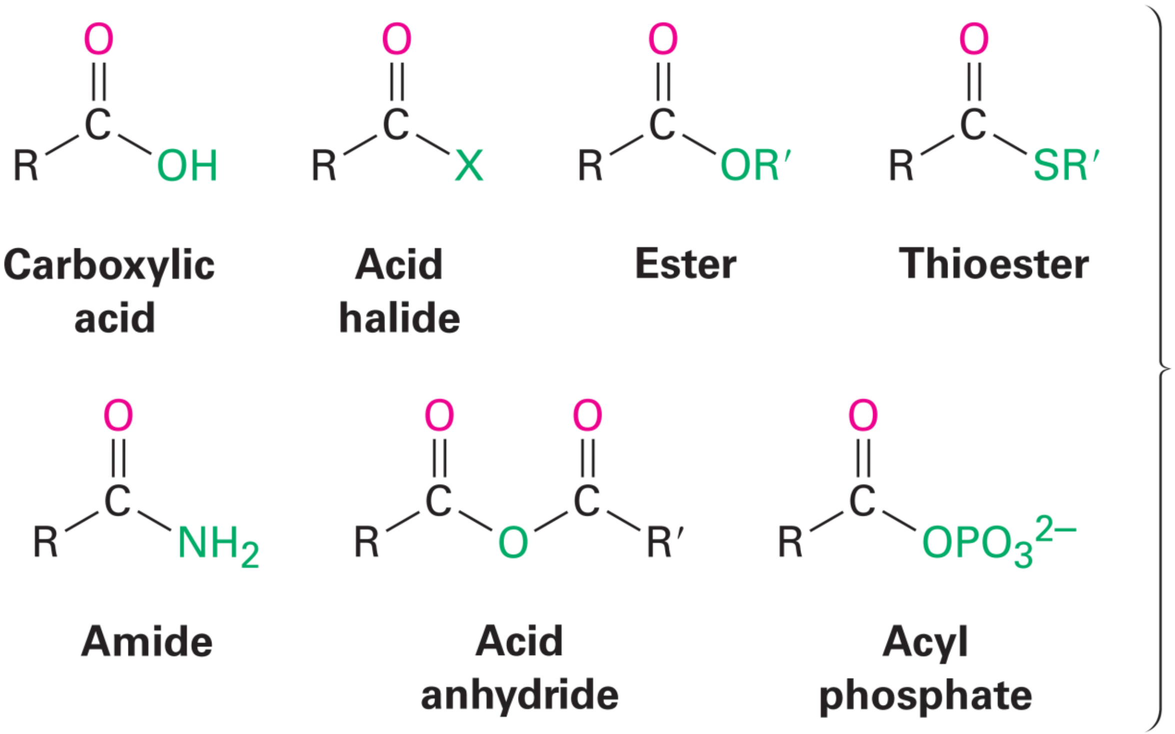
CHƯƠNG 11 – CARBOXYLIC ACIDS & CÁC DẪN XUẤT



Tham gia phản ứng thế ái nhân khi -OH, -X, -OR... có thể tách ra thành những ion bền

Ref. John McMurry, *Organic Chemistry - Eighth Edition*, 2010, Cengage Learning Publication

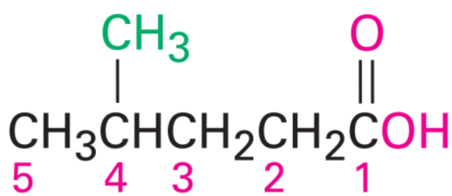
Summary of Functional Group Nomenclature

Class	Suffix Name	Prefix Name
Carboxylic acid	-oic acid	Carboxy
Ester	-oate	Alkoxycarbonyl
Amide	-amide	Amido
Nitrile	-nitrile	Cyano
Aldehyde	-al	Oxo (=O)
Aldehyde	-al	Formyl (—CH=O)
Ketone	-one	Oxo (=O)
Alcohol	-ol	Hydroxy
Amine	-amine	Amino
Alkene	-ene	Alkenyl
Alkyne	-yne	Alkynyl
Alkane	-ane	Alkyl
Ether	—	Alkoxy
Alkyl halide	—	Halo

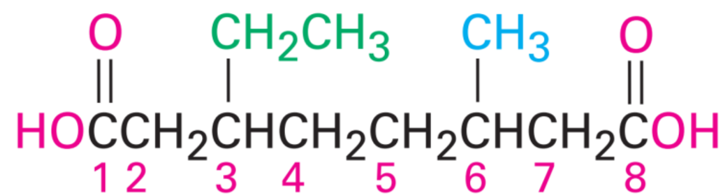
increasing priority

ĐỌC TÊN ACID

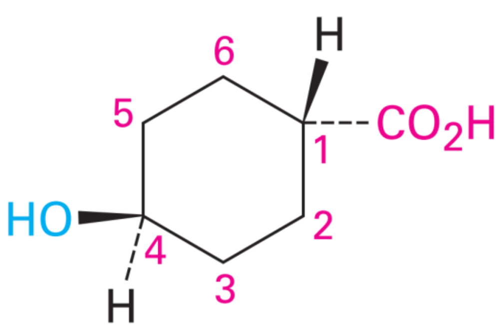
- Danh pháp IUPAC cho acid mạch hở: **hydrocarbon+oic acid** (methanoic acid, propanoic…)
- Danh pháp IUPAC cho acid có nhóm –COOH gắn trực tiếp lên vòng: tên vòng+carboxylic acid (cyclohexanecarboxylic acid…)



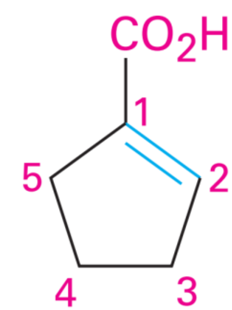
4-Methylpentanoic acid



3-Ethyl-6-methyloctanedioic acid



***trans*-4-Hydroxycyclohexanecarboxylic acid**



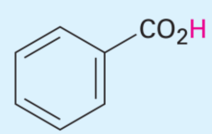
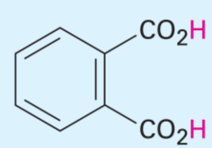
1-Cyclopentenecarboxylic acid

Ref. John McMurry, *Organic Chemistry - Eighth Edition*, 2010, Cengage Learning Publication

ĐỌC TÊN ACID

- Tên thông dụng của một số acid

Structure	Name	Acyl group
HCO_2H	Formic	Formyl
CH_3CO_2H	Acetic	Acetyl
$CH_3CH_2CO_2H$	Propionic	Propionyl
$CH_3CH_2CH_2CO_2H$	Butyric	Butyryl
HO_2CCO_2H	Oxalic	Oxalyl
$HO_2CCH_2CO_2H$	Malonic	Malonyl
$HO_2CCH_2CH_2CO_2H$	Succinic	Succinyl
$HO_2CCH_2CH_2CH_2CO_2H$	Glutaric	Glutaryl
$HO_2CCH_2CH_2CH_2CH_2CO_2H$	Adipic	Adipoyl
$H_2C{=}CHCO_2H$	Acrylic	Acryloyl
$HO_2CCH{=}CHCO_2H$	Maleic (cis)	Maleoyl
	Fumaric (trans)	Fumaroyl
$HOCH_2CO_2H$	Glycolic	Glycoloyl
$CH_3\overset{OH}{\overset{\vert}{C}}HCO_2H$	Lactic	Lactoyl

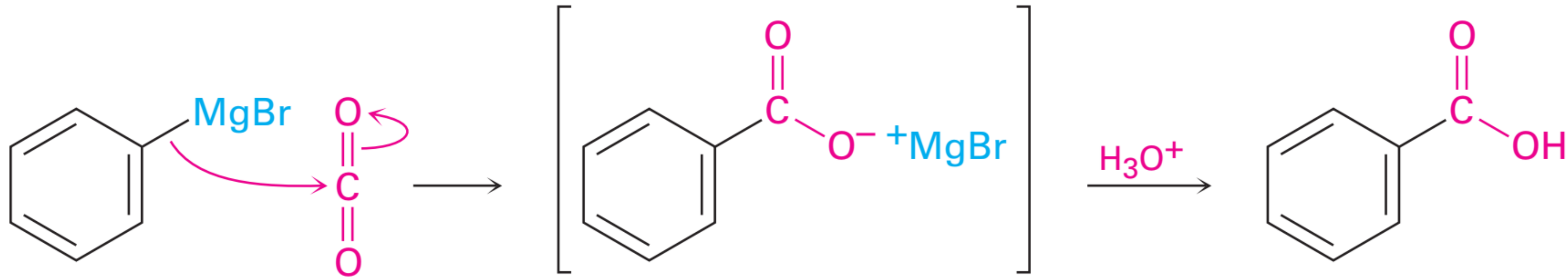
Structure	Name	Acyl group
$CH_3\overset{O}{\overset{\Vert}{C}}CO_2H$	Pyruvic	Pyruvoyl
$HOCH_2\overset{OH}{\overset{\vert}{C}}HCO_2H$	Glyceric	Gylceroyl
$HO_2C\overset{OH}{\overset{\vert}{C}}HCH_2CO_2H$	Malic	Maloyl
$HO_2C\overset{O}{\overset{\Vert}{C}}CH_2CO_2H$	Oxaloacetic	Oxaloacetyl
	Benzoic	Benzoyl
	Phthalic	Phthaloyl

Ref. John McMurry, *Organic Chemistry - Eighth Edition*, 2010, Cengage Learning Publication

ĐIỀU CHẾ CARBOXYLIC ACID

Từ hợp chất cơ magnesium RMgX

- 1. Tác dụng với CO_2 (đây là hợp chất vô cơ), 2. Thủy phân
- Acid tăng 1 C (của –COOH) so với tác chất ban đầu

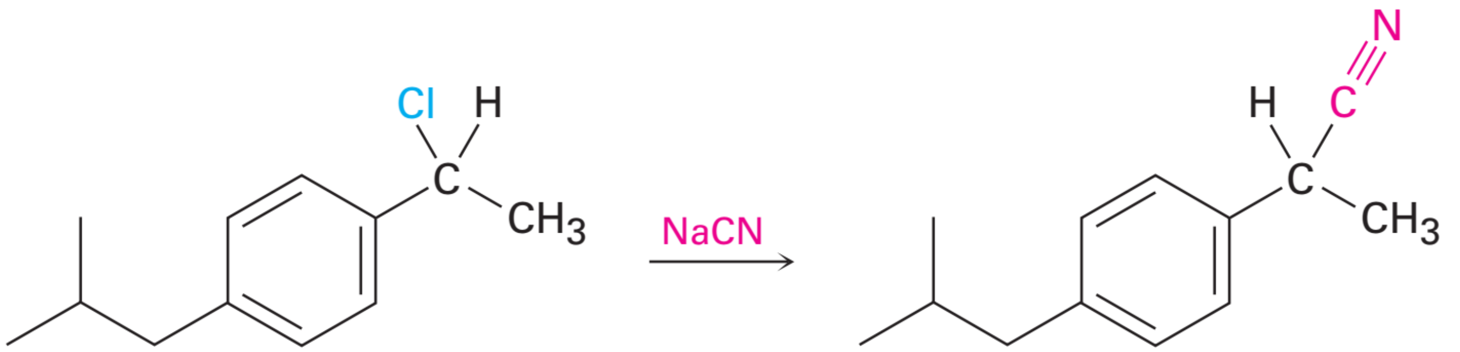


Phenylmagnesium bromide

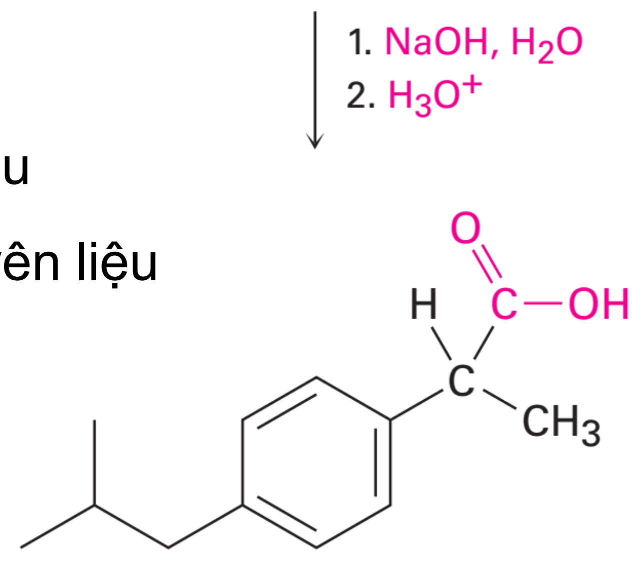
Benzoic acid

ĐIỀU CHẾ CARBOXYLIC ACID

Từ hợp chất nitrile



- Thủy phân trong môi trường acid
- Acid có số C không đổi so với tác chất ban đầu
- Lưu ý phương pháp đưa nhóm –CN vào nguyên liệu

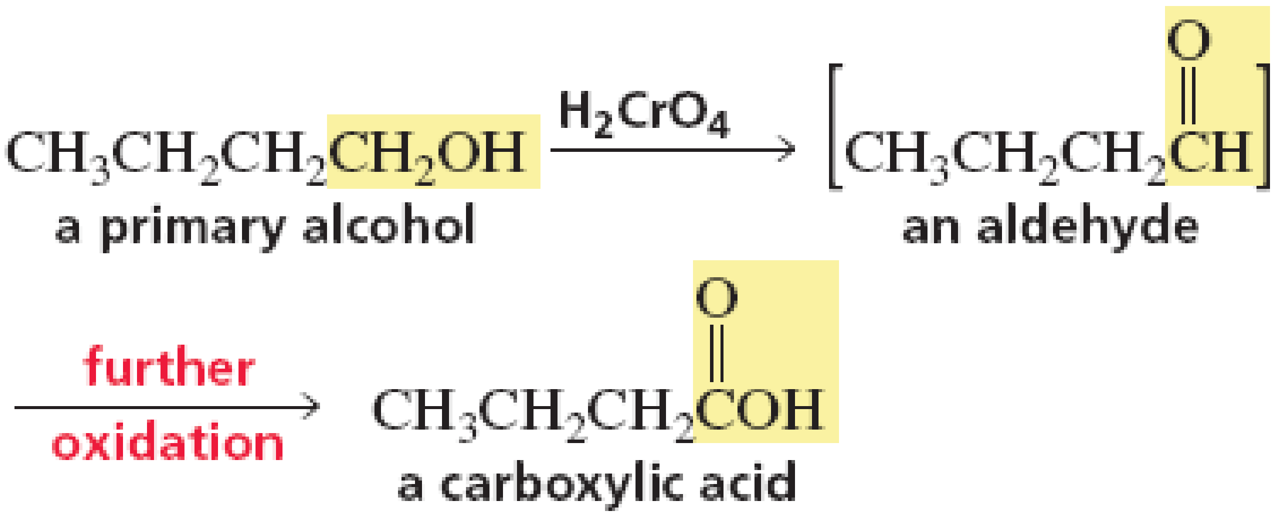


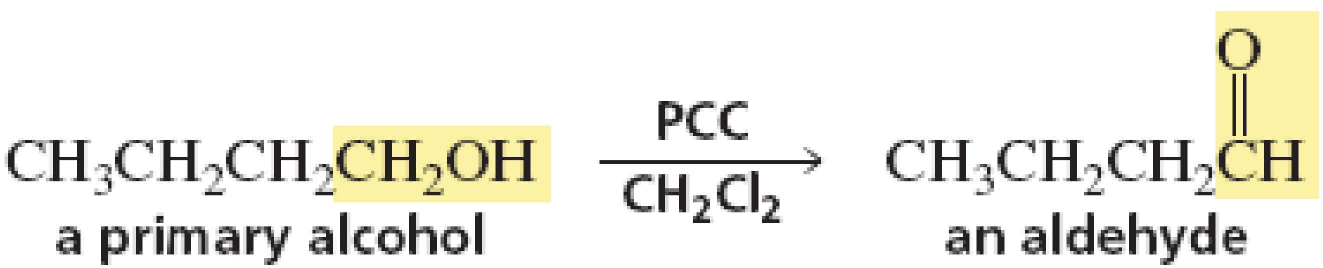
Ibuprofen

ĐIỀU CHẾ CARBOXYLIC ACID

Từ alcohol bậc 1

- Oxy hóa alcohol bậc 1 không dừng lại ở aldehyde
- Ngoại trừ tác nhân PCC oxi hóa êm dịu đến aldehyde

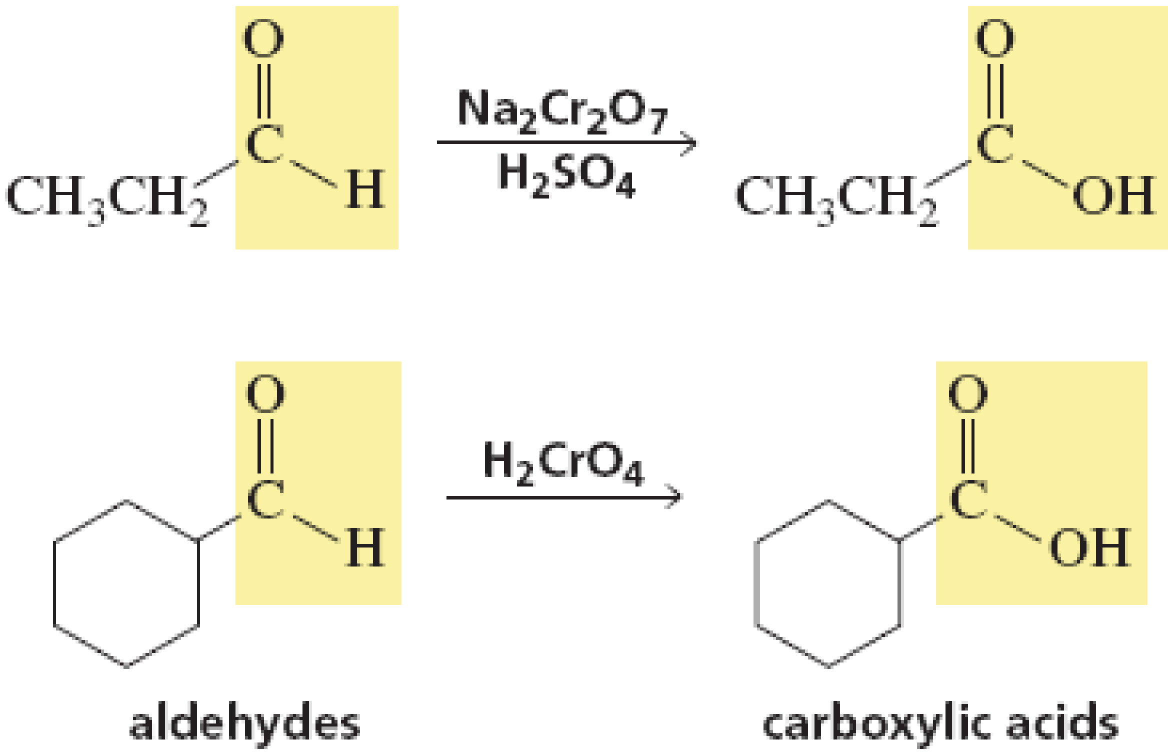




ĐIỀU CHẾ CARBOXYLIC ACID

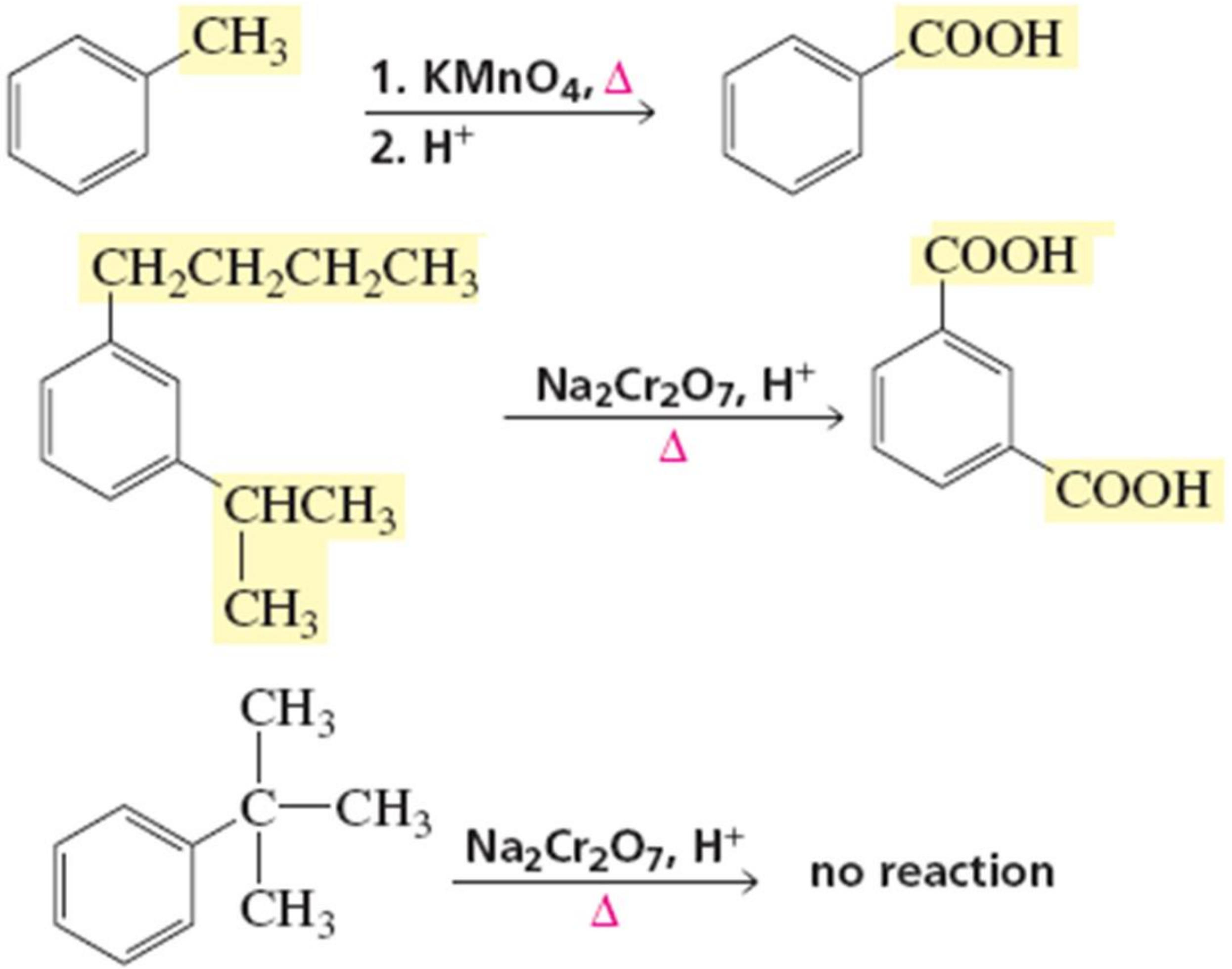
Từ aldehyde

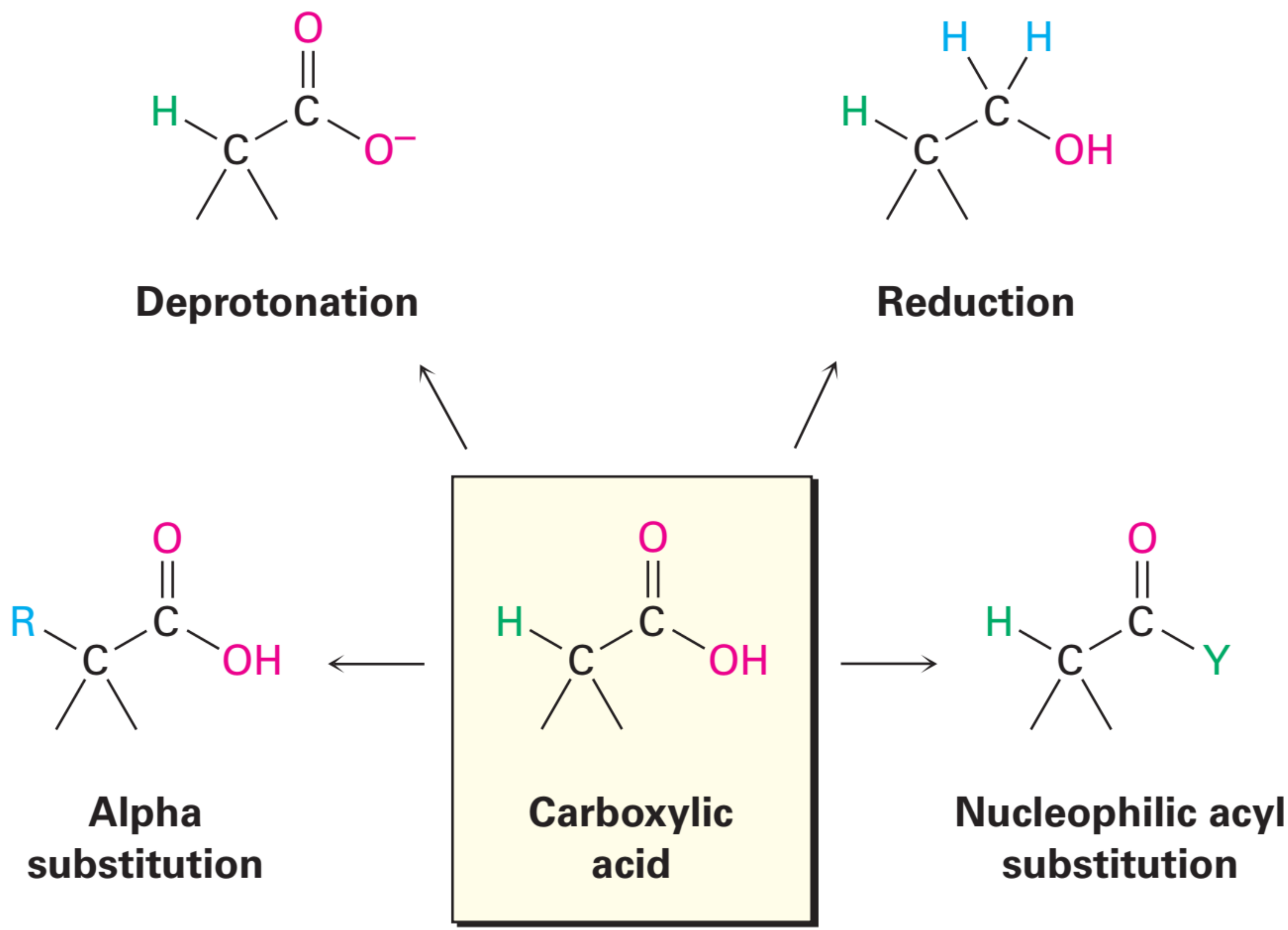
- Rất không bền, dễ bị oxi hóa thành acid khi có mặt tác nhân oxi hóa



ĐIỀU CHẾ CARBOXYLIC ACID

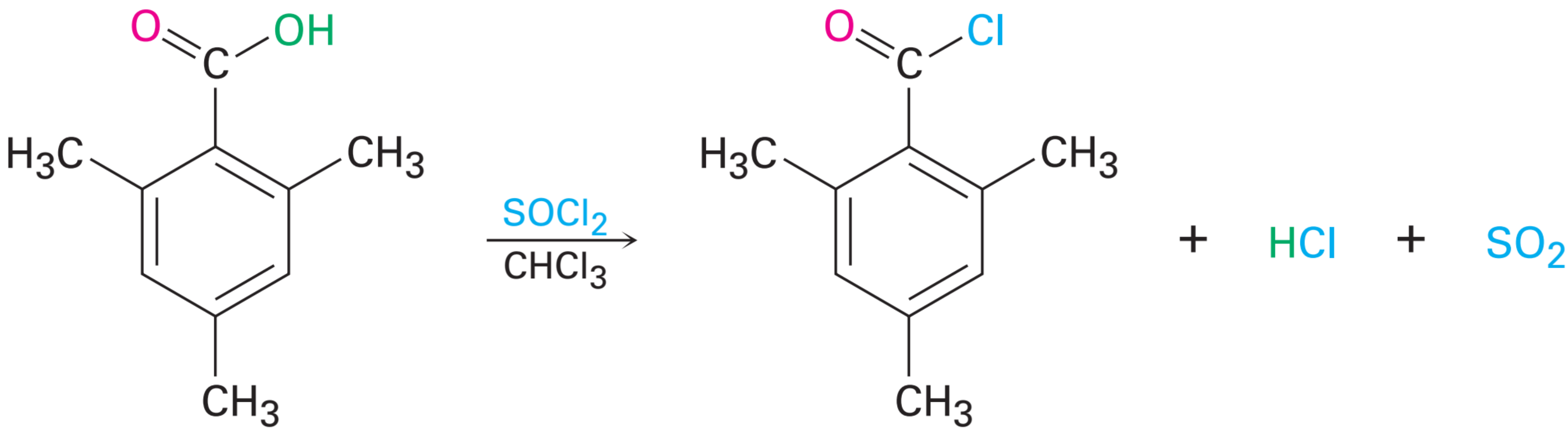
Từ alkylbenzene




Deprotonation
Reduction
Alpha substitution
Carboxylic acid
Nucleophilic acyl substitution

PHẢN ỨNG THẾ VỚI $SOCl_2$/PCl_n/PBr_n(n=3,5)

Điều chế ACYL HALIDE (RCOX)



2,4,6-Trimethylbenzoic acid

2,4,6-Trimethylbenzoyl chloride (90%)

PHẢN ỨNG ESTER HÓA VỚI ALCOHOL

$$\underset{\text{Alcohol}}{ROH} + \underset{\text{Carboxylic acid}}{R'\overset{O}{\overset{\|}{C}}OH} \xrightleftharpoons{H^+} \underset{\text{Ester}}{R'\overset{O}{\overset{\|}{C}}OR} + \underset{\text{Water}}{H_2O}$$

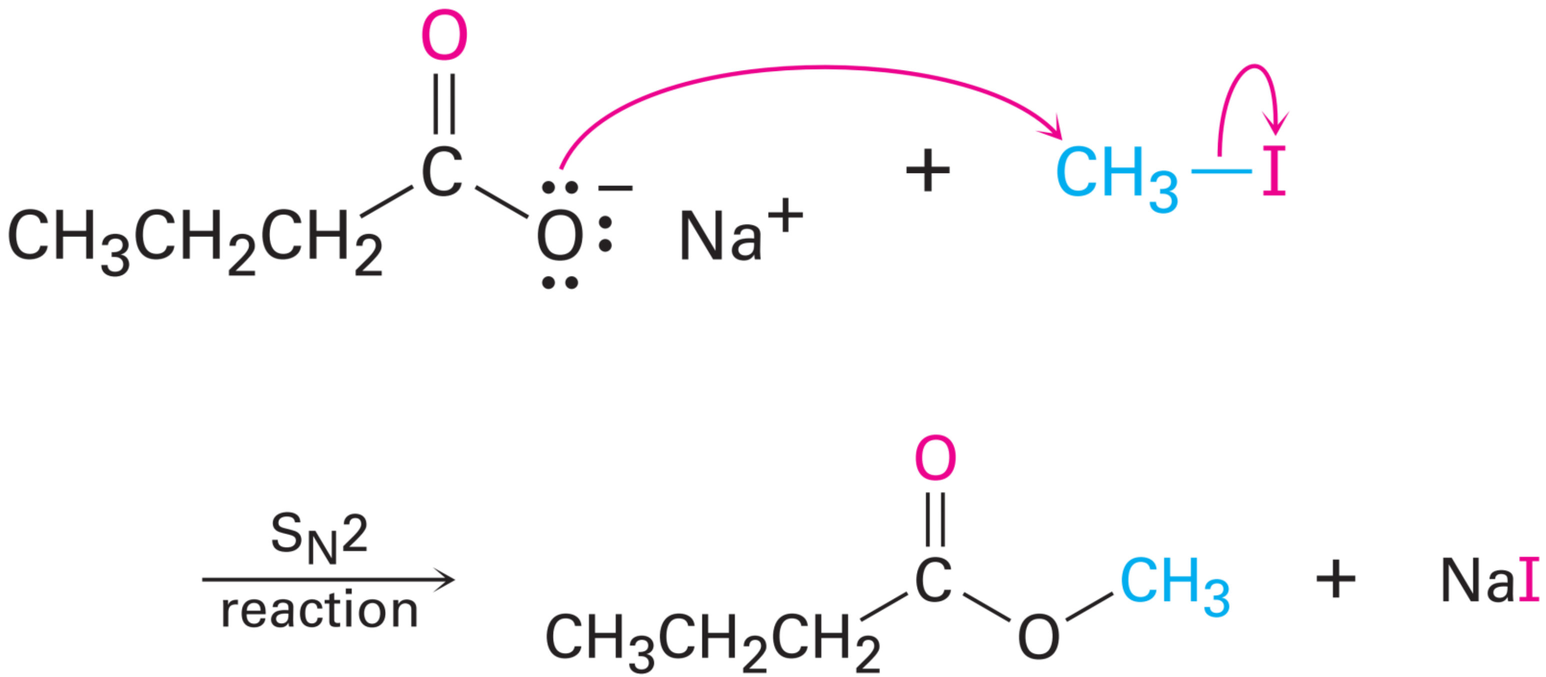
$$\underset{\text{Alcohol}}{ROH} + \underset{\text{Carboxylic acid anhydride}}{R'\overset{O}{\overset{\|}{C}}O\overset{O}{\overset{\|}{C}}R'} \longrightarrow \underset{\text{Ester}}{R'\overset{O}{\overset{\|}{C}}OR} + \underset{\text{Carboxylic acid}}{R'\overset{O}{\overset{\|}{C}}OH}$$

$$\underset{\text{Alcohol}}{ROH} + \underset{\text{Acyl chloride}}{R'\overset{O}{\overset{\|}{C}}Cl} \longrightarrow \underset{\text{Ester}}{R'\overset{O}{\overset{\|}{C}}OR} + \underset{\text{Hydrogen chloride}}{HCl}$$

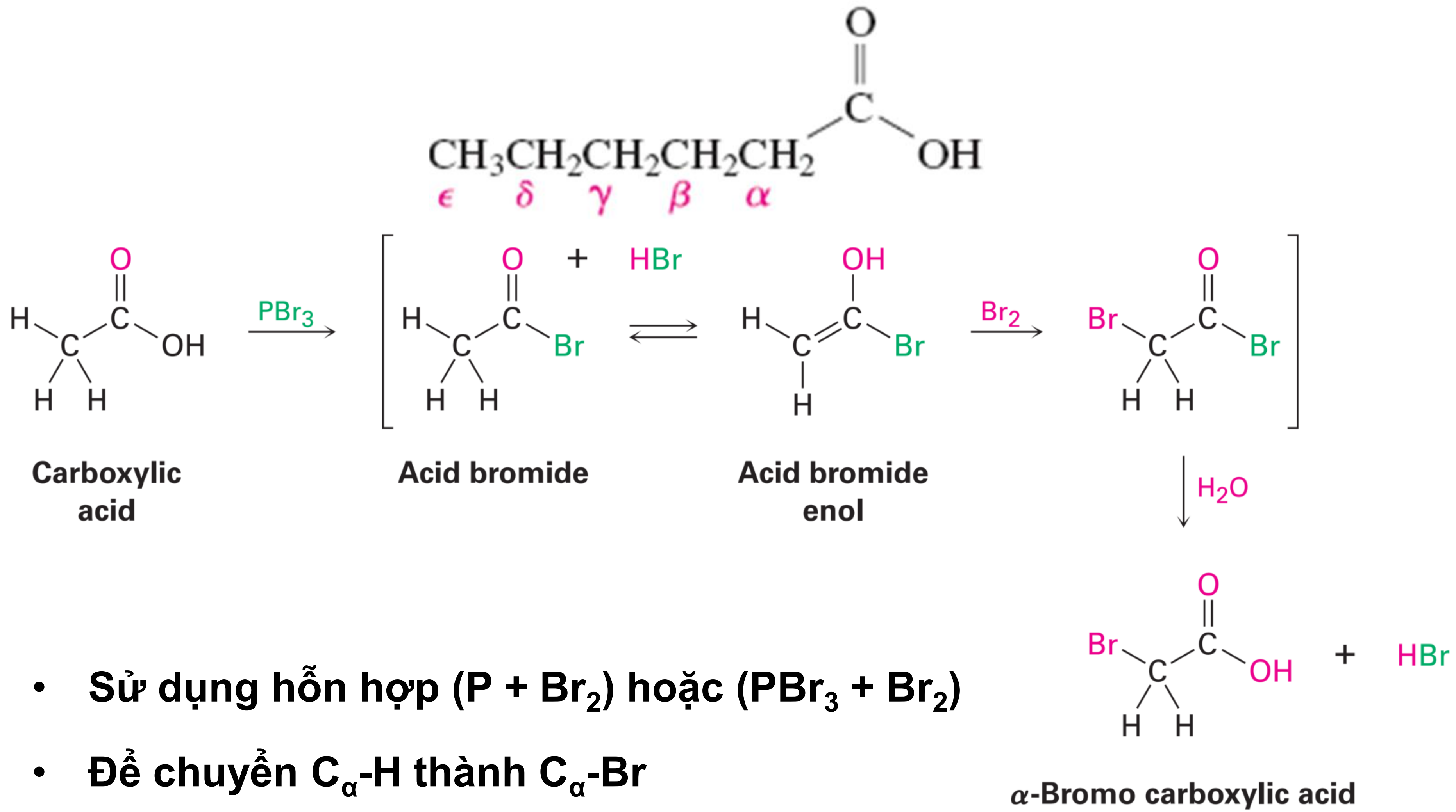
- **Phenol không phản ứng với acid để tạo ester**

PHẢN ỨNG ESTER HÓA VỚI DẪN XUẤT HALOGEN

- **Từ muối carboxylate (tác nhân S_N2)**



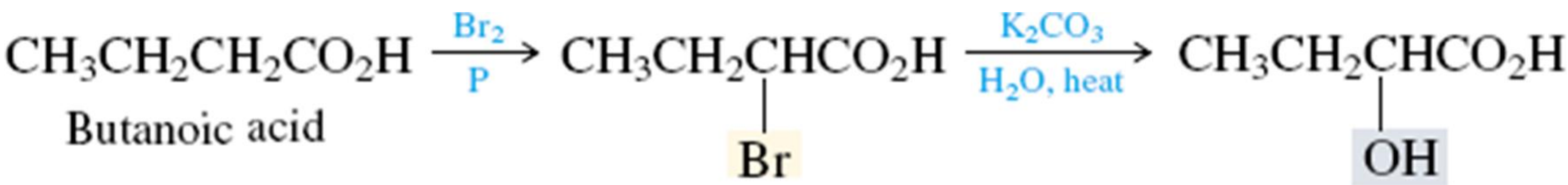
PHẢN ỨNG BROMO HÓA VỚI TẠI C_α

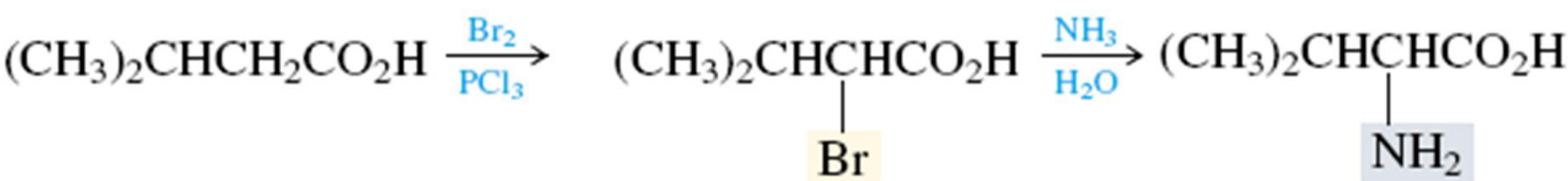


- Sử dụng hỗn hợp (P + Br_2) hoặc (PBr_3 + Br_2)
- Để chuyển C_α-H thành C_α-Br

Từ α-bromo carboxylic acid

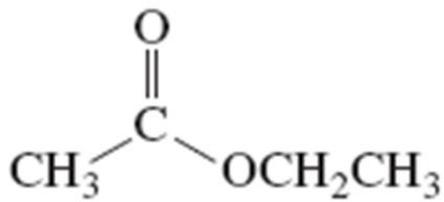
- **Dễ dàng chuyển hóa thành những nhóm thế khác như -OH, -NH$_2$, -CN bên cạnh -COOH**

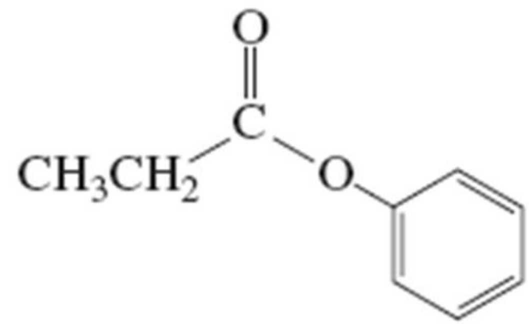




Đọc tên ester RCOOR’

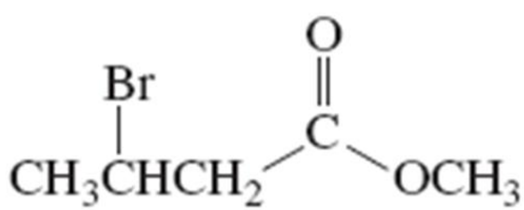
Tên mạch alkyl của gốc rượu (R’) + tên gốc acid (chuyển *-ic acid* thành *-ate*)

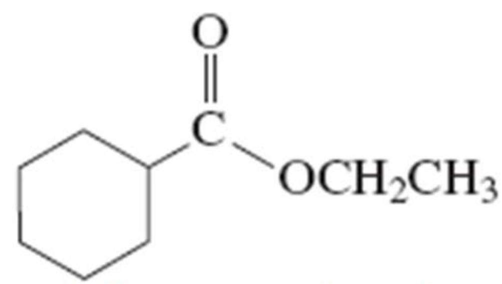




Tên IUPAC: ethyl ethanoate — phenyl propanoate

Tên thông dụng: ethyl acetate — phenyl propionate



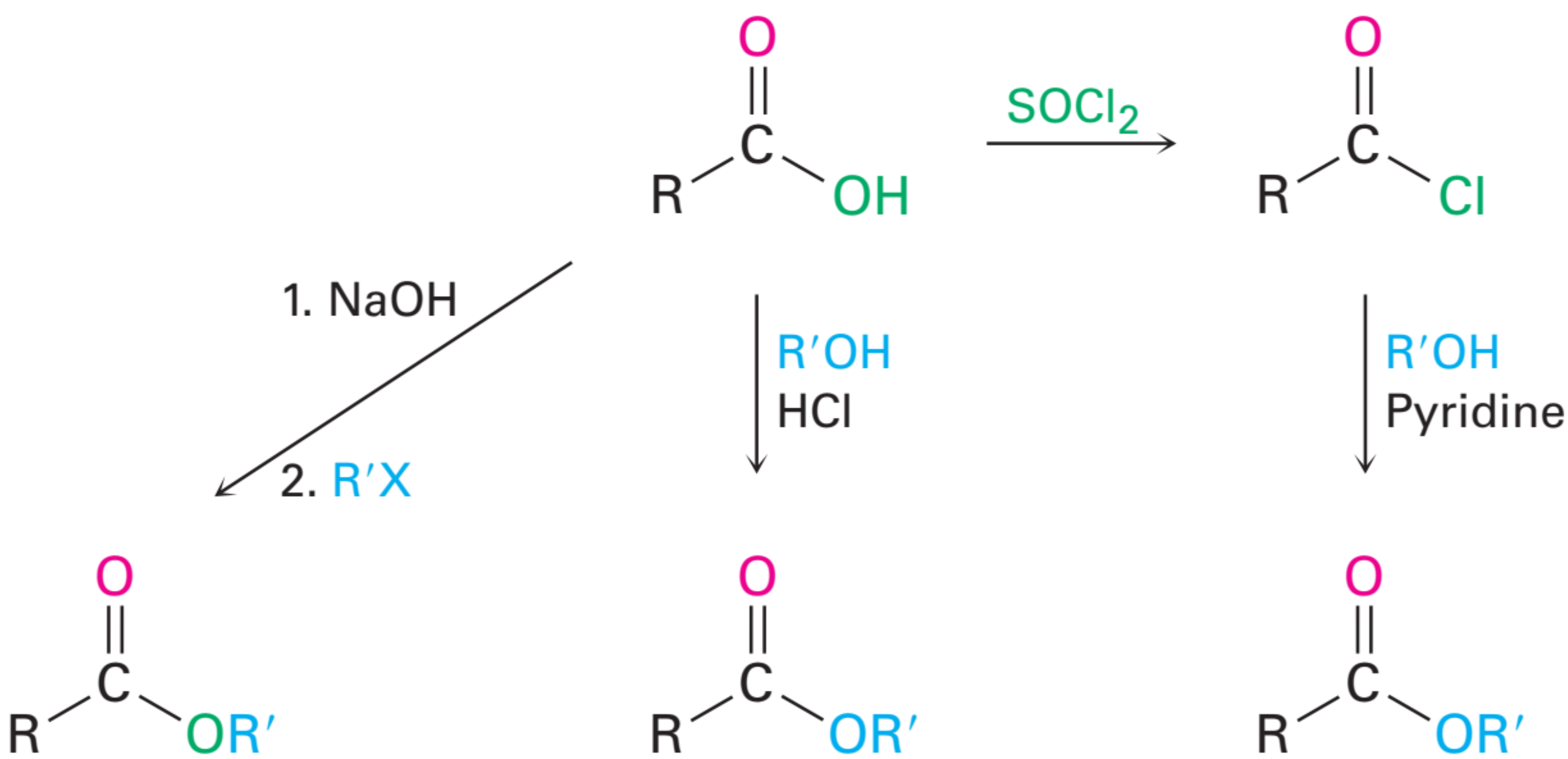


methyl 3-bromobutanoate
methyl β-bromobutyrate

ethyl cyclohexanecarboxylate

Ref. John McMurry, *Organic Chemistry - Eighth Edition*, 2010, Cengage Learning Publication

Điều chế ester

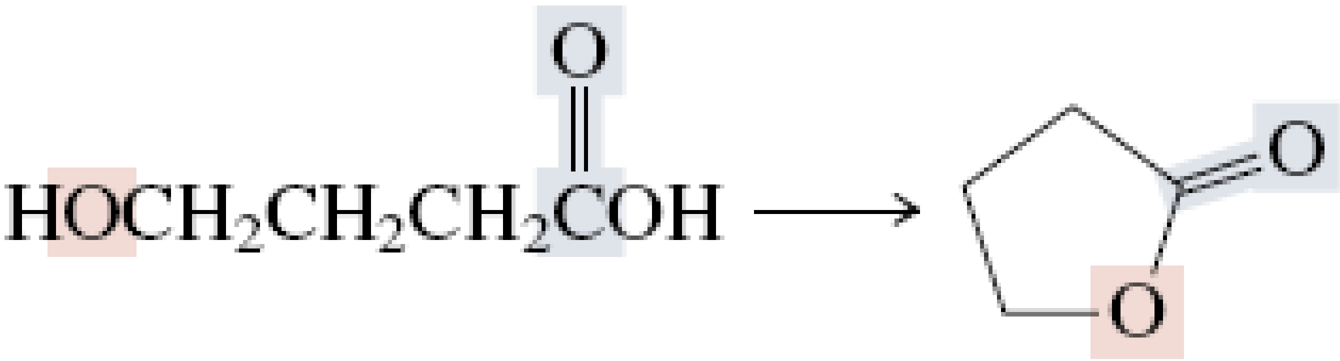


Chỉ áp dụng cho R'X bậc 1

Đơn giản nhất
Thuận nghịch

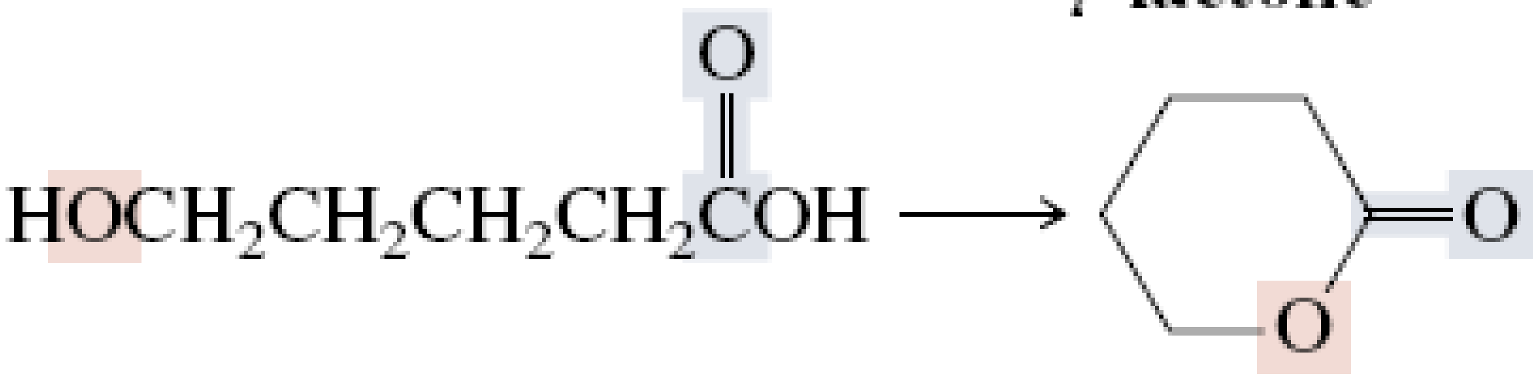
Tốn kém
Hiệu suất cao

Phản ứng tạo lactone vòng



4-Hydroxybutanoic acid

4-Butanolide
γ-lactone



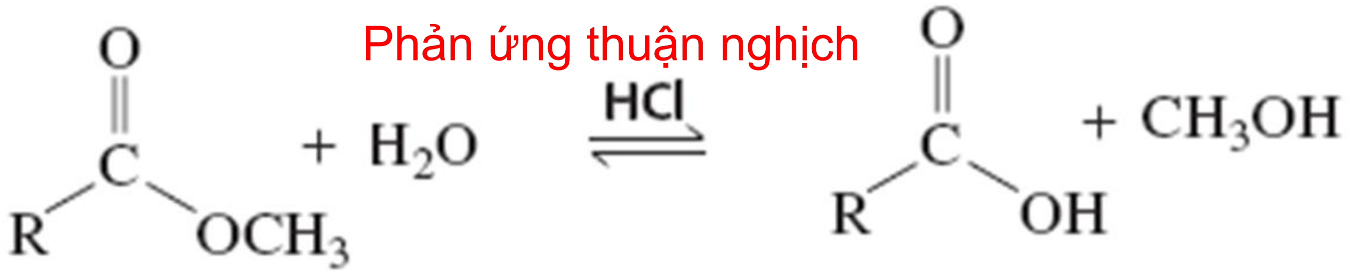
5-Hydroxypentanoic acid

5-Pentanolide

δ-lactone

Phản ứng thủy phân ester

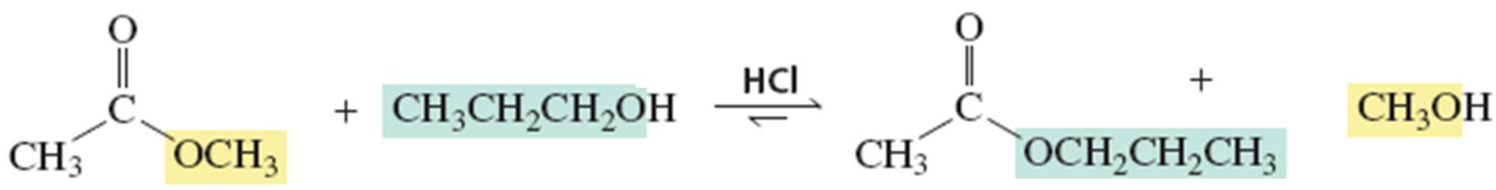
Phản ứng thuận nghịch

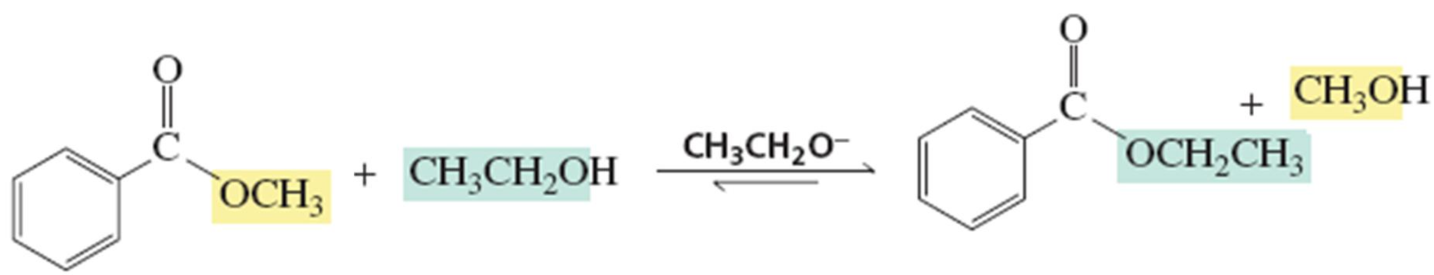


Phản ứng một chiều

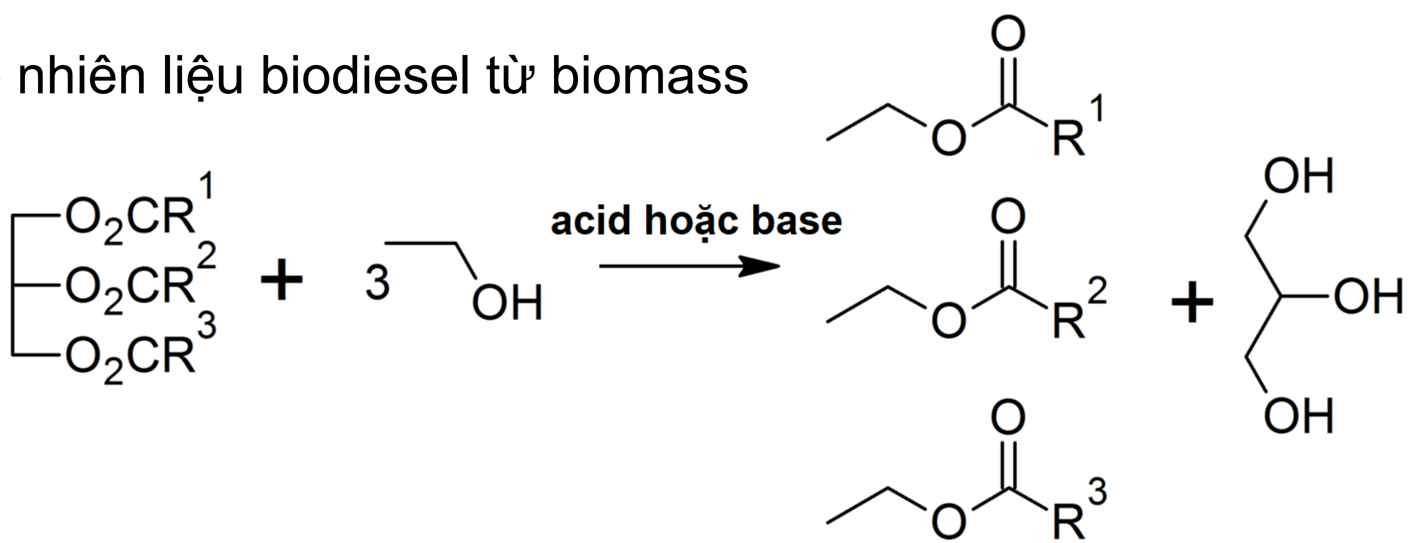
Phản ứng trao đổi ester

- Trao đổi gốc rượu tạo thành ester mới trong môi trường acid hoặc base



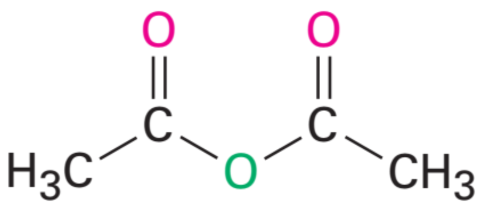


- Tổng hợp nhiên liệu biodiesel từ biomass

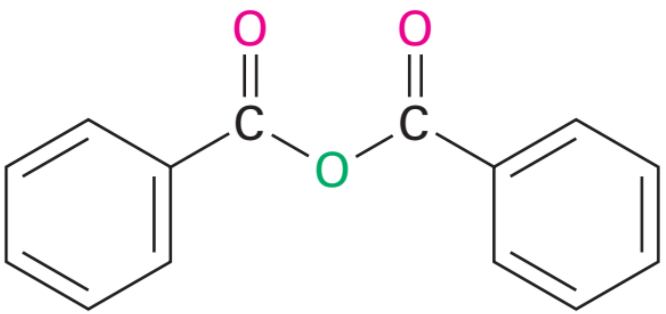


Đọc tên acid anhydride $(RCO_2)O$

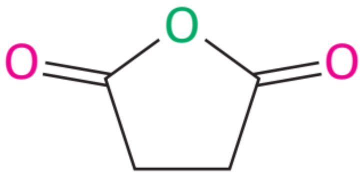
Tên acid (bỏ đuôi *acid*) + anhydride



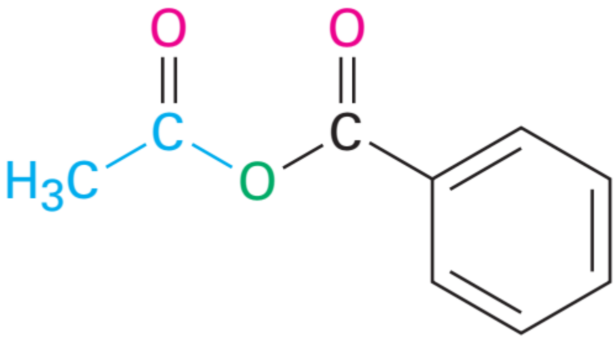
Acetic anhydride



Benzoic anhydride



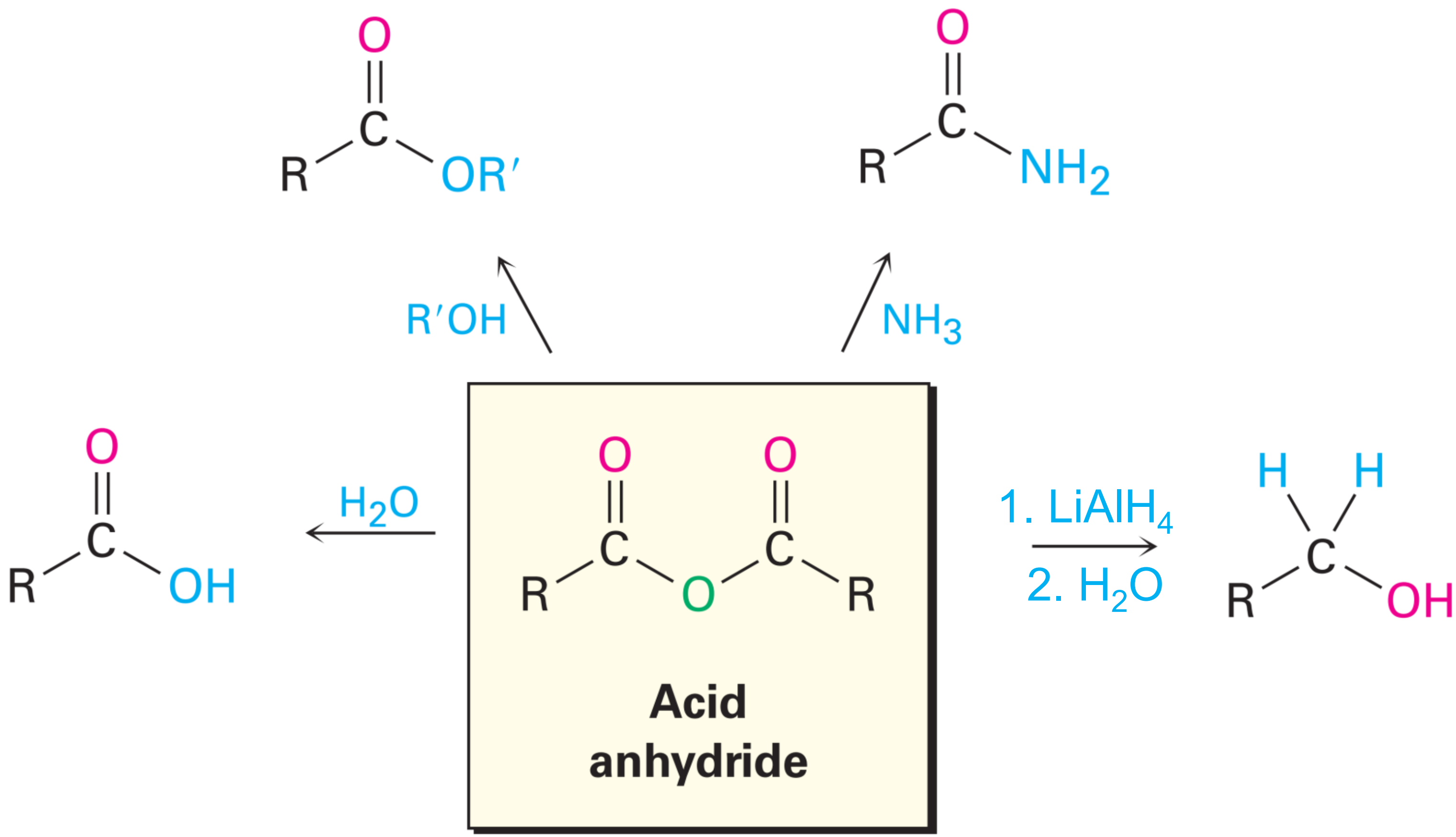
Succinic anhydride



Acetic benzoic anhydride

Ref. John McMurry, *Organic Chemistry - Eighth Edition*, 2010, Cengage Learning Publication

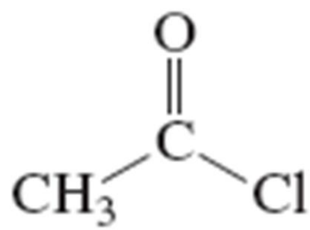
Phản ứng của acid anhydride $(RCO_2)O$

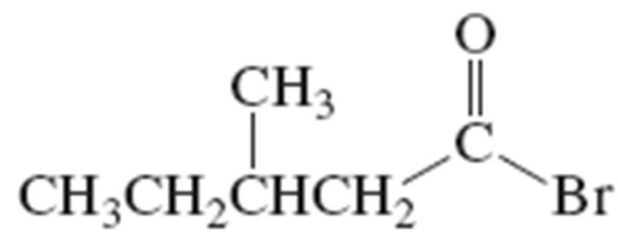


Ref. John McMurry, *Organic Chemistry - Eighth Edition*, 2010, Cengage Learning Publication

Đọc tên acid halide RCOX

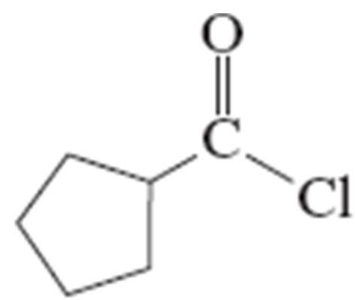
Tên gốc acid (chuyển *-ic acid* thành *-yl*) + halide





Tên IUPAC: ethanoyl chloride | 3-methylpentanoyl bromide

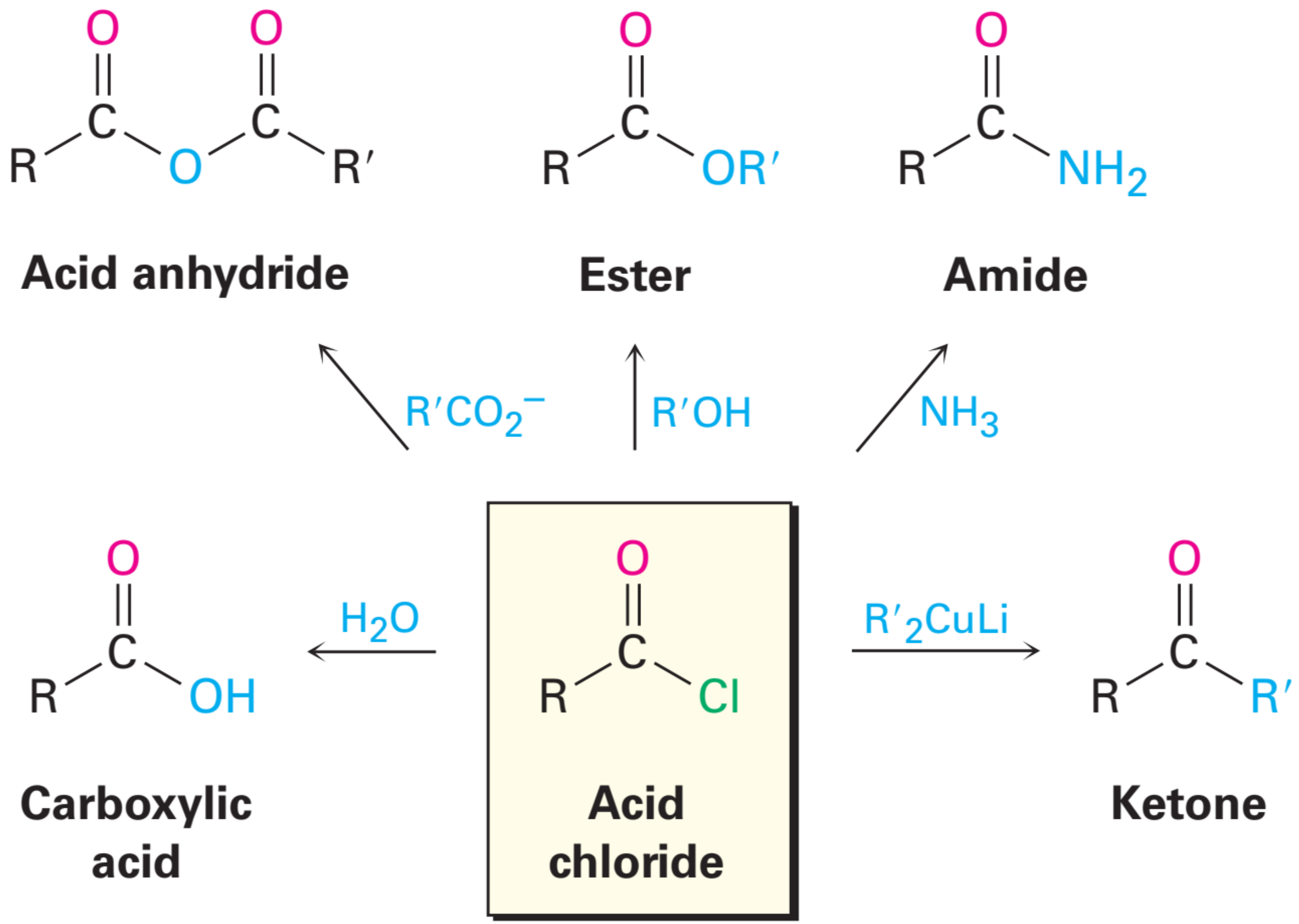
Tên thông dụng: acetyl chloride | β-methylvaleryl bromide



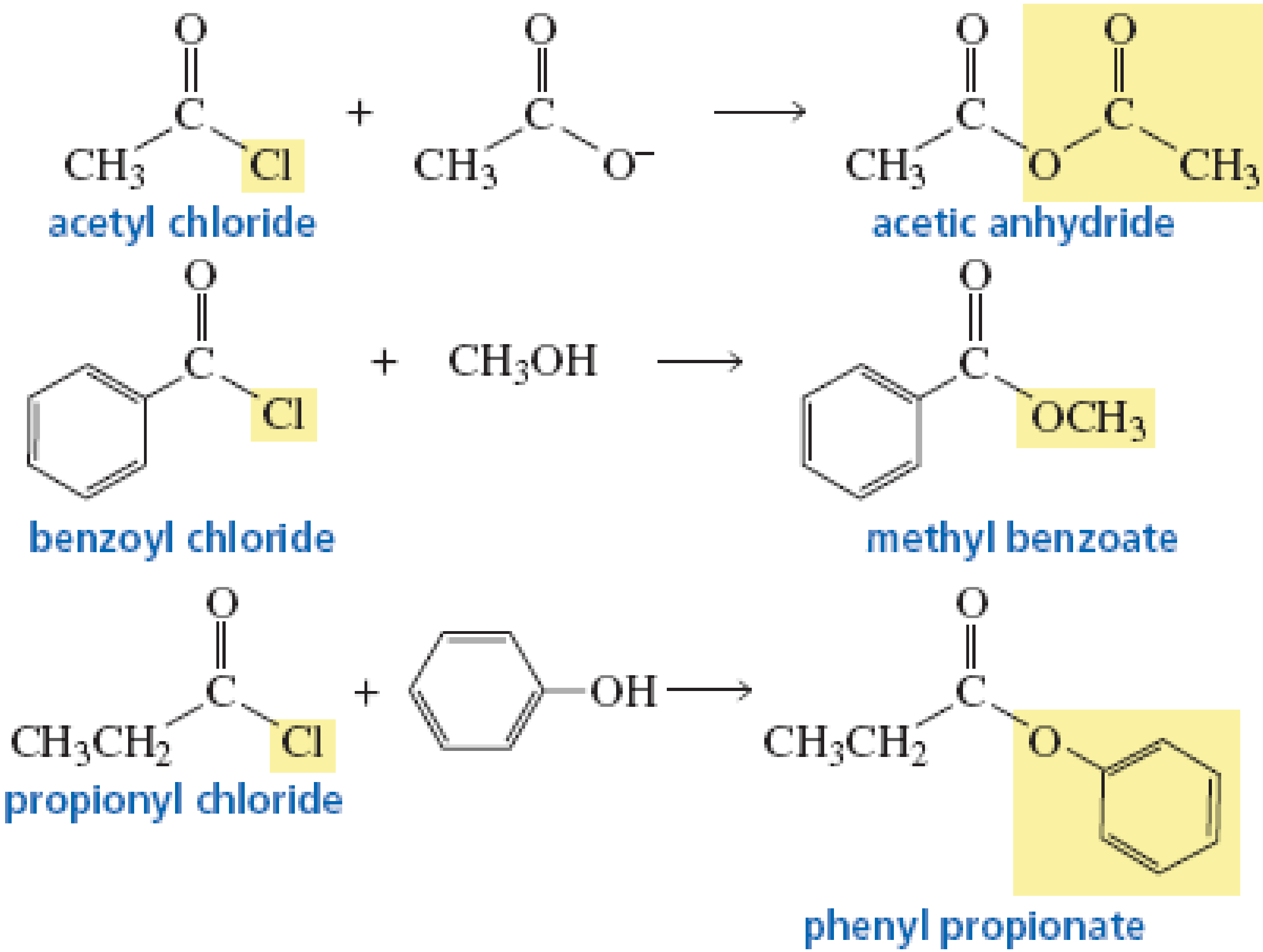
cyclopentanecarbonyl chloride

Ref. John McMurry, *Organic Chemistry - Eighth Edition*, 2010, Cengage Learning Publication

Phản ứng tiêu biểu của acyl halide



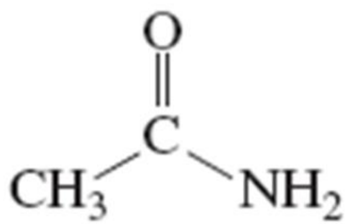
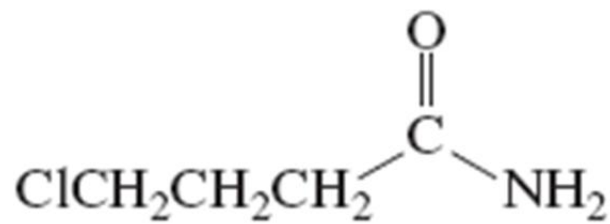
Phản ứng tiêu biểu của acyl halide



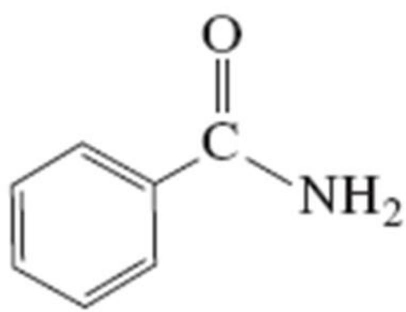
Ref. John McMurry, *Organic Chemistry - Eighth Edition*, 2010, Cengage Learning Publication

Đọc tên amide RCONHR’

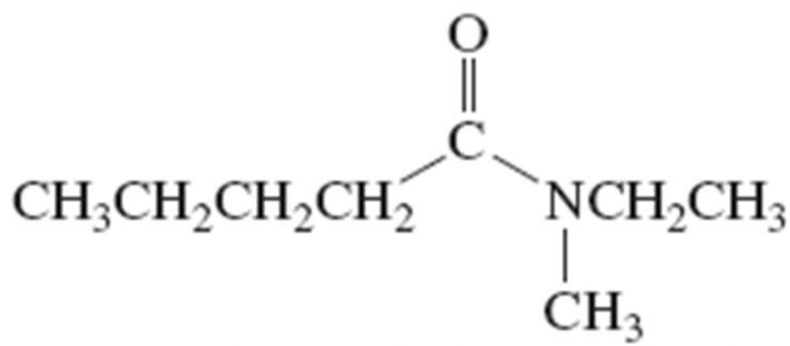
Chuyển *-ic acid/-xylic acid/-oic acid* thành *amide*

Tên IUPAC:	ethanamide	4-chlorobutanamide
Tên thông dụng:	acetamide	γ-chlorobutyramide



benzenecarboxamide
benzamide



N-ethyl-*N*-methylpentanamide

Ref. John McMurry, *Organic Chemistry - Eighth Edition*, 2010, Cengage Learning Publication

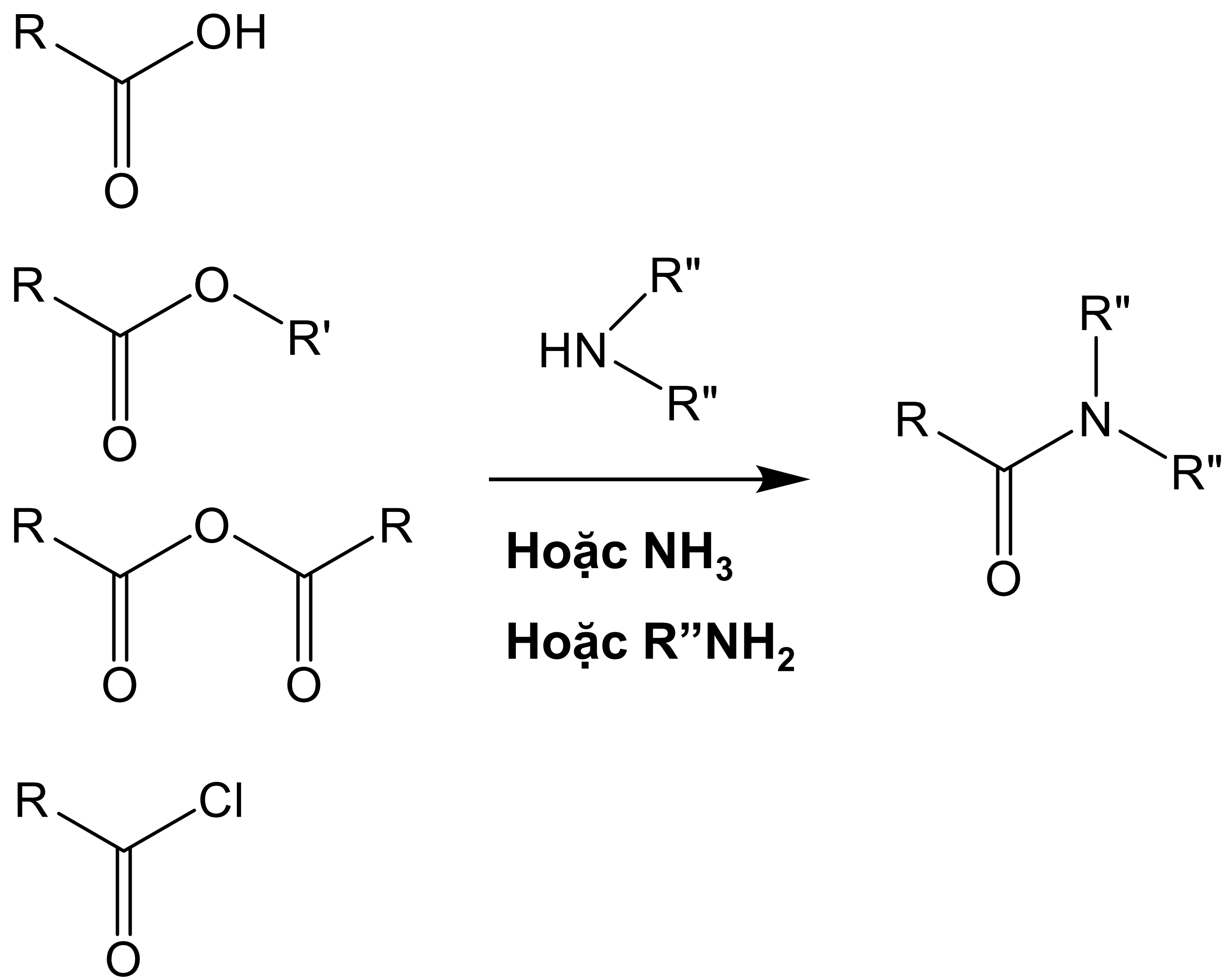
Điều chế amide

Acid

Ester

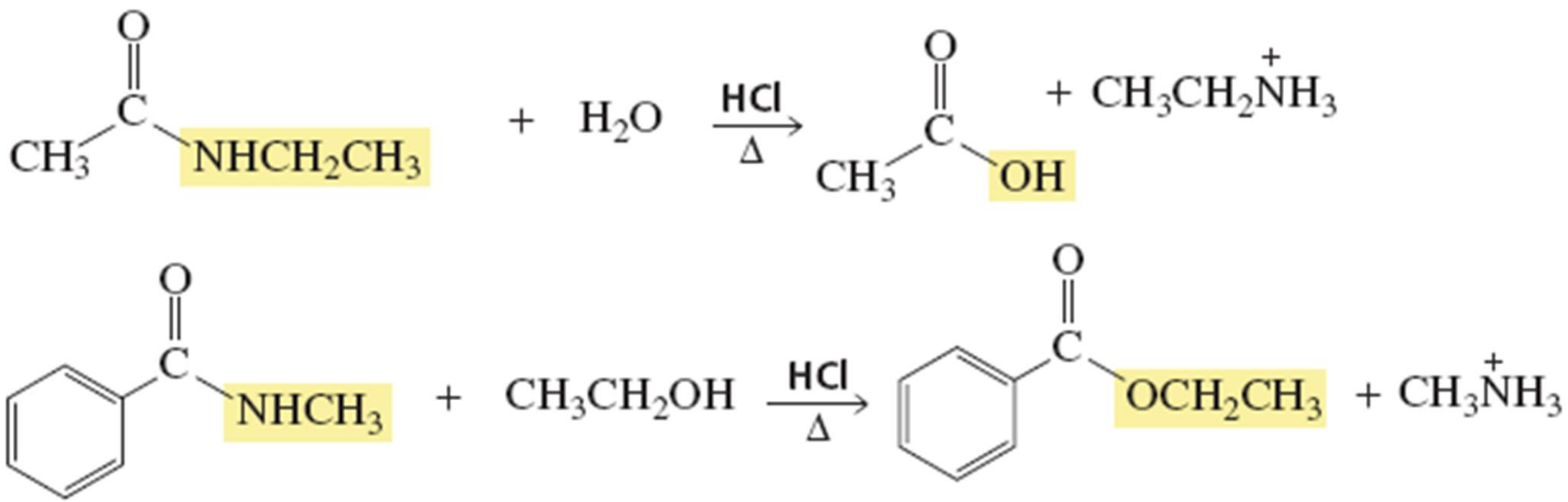
Anhydride

Acyl halide



Ref. John McMurry, *Organic Chemistry - Eighth Edition*, 2010, Cengage Learning Publication

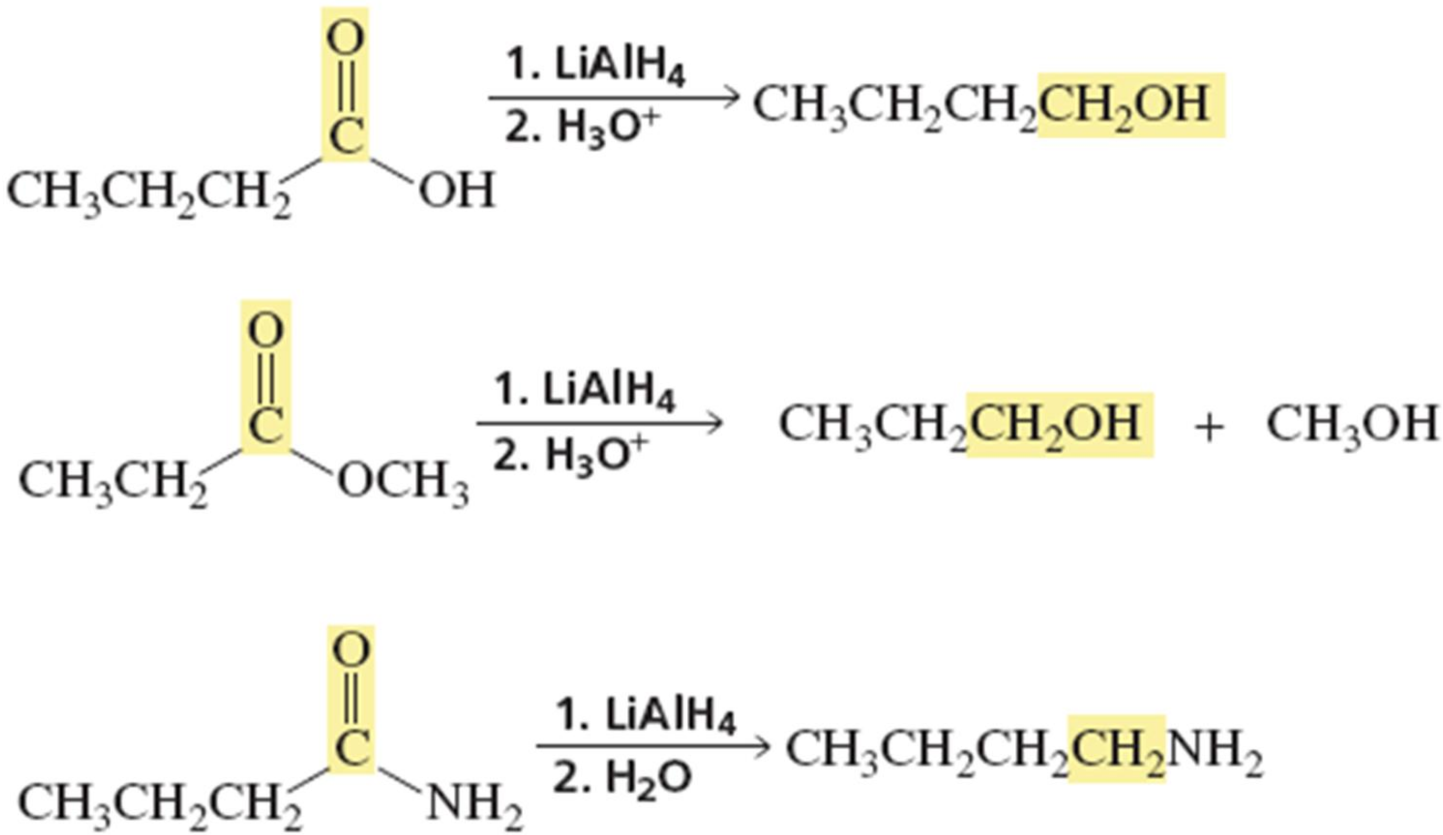
Phản ứng của amide



Ref. John McMurry, *Organic Chemistry - Eighth Edition*, 2010, Cengage Learning Publication

PHẢN ỨNG KHỬ

Acid, ester và amide bị khử thành rượu hoặc amine bằng $LiAlH_4$



Ref. John McMurry, *Organic Chemistry - Eighth Edition*, 2010, Cengage Learning Publication